

NUR AHMAD KHATIM

Jakarta, Indonesia

Phone: +62 852-5566-0655

LinkedIn: [linkedin.com/in/nurahmadkhatim](https://www.linkedin.com/in/nurahmadkhatim)

Email: nurahmadkhatim@gmail.com

Web: nurahmadkhatim.github.io

SUMMARY

Software engineer and AI researcher with 3+ years of full-stack development experience and a growing publication record in explainable AI and decision-making under uncertainty. Builds production systems across fintech, HR, and education domains while contributing to peer-reviewed research with Stanford SISL and KFUPM.

EDUCATION

Bachelor of Informatics

2019 – 2023

Sepuluh Nopember Institute of Technology (ITS), Surabaya, Indonesia

GPA: 3.73/4.00

PROFESSIONAL EXPERIENCE

Software Engineer

2025 – Present

Ewakopedia (PT. Ewakopedia Santri Teknologi), Indonesia

- Developed smart system cashless payment and comprehensive reporting for boarding school
- Developed membership, order tracking, rewarding, and SEO-friendly article management system
- Developed scientific journal publication management system

Software Maintenance

2023 – 2025

DataOn (PT. Indodev Niaga Internet), South Tangerang, Indonesia

- Supported 21 client projects using the company's HRIS application while maintaining high SLA compliance
- Resolved approximately 25 software issues per month through debugging, code fixes, and close collaboration with various internal teams

Software Engineer

2022 – 2023

Suitmedia (PT. Suitmedia Kreasi Indonesia), Indonesia

- Implemented new features and performed existing features maintenance of stock exchange trading operation support system
- Improved system reliability and scalability by optimizing code and conducting regular reviews

Teaching Assistant

2021 – 2022

Sepuluh Nopember Institute of Technology (ITS), Surabaya, Indonesia

- Assisted in teaching a Data Structure course to 72 students
- Conducted bi-weekly lab sessions and practice exercises to reinforce course material and foster hands-on learning

RESEARCH EXPERIENCE

Research Assistant

April 2025 – Present

AI Verification & Validation Lab (AI V&V), KFUPM, Remote

- Co-authored research on an explainable AI (XAI) framework for automatic Kellgren-Lawrence grading of knee osteoarthritis, contributing to model development and the resulting publication (Knee-xRAI)
- Developed an optimization model for human-robot co-dispatch planning in petroleum-site surveillance under varying task criticalities, resulting in a published paper (HRCD-FLP)
- Mentored 3 research interns, providing technical guidance on model development, experimentation, and research methodology

Research Assistant

Oct 2023 – Jan 2024

Stanford Intelligent Systems Laboratory (SISL), Stanford University, Remote

- Developed an integrated decision making under uncertainty model (POMDP) for healthcare
- Collaborated with a multidisciplinary team of faculty, clinicians, and graduate researchers

Research Assistant

Jan 2023 – Aug 2023

Algorithm and Programming Laboratory (AP) ITS, Surabaya, Indonesia

- Developed a machine learning model based on 1D-CNN for temporal signal
- Conducted feature engineering on real-world datasets, improving model accuracy by 20%

PUBLICATIONS

- Irfan, Azmul A., **Khatim, N. A.**, Irfan, Alfian. A., Zaki, A., Suwarsono, E. A., & Arief, M. M. (2026). Knee-xRAI: An Explainable AI Framework for Automatic Kellgren-Lawrence Grading of Knee Osteoarthritis. *arXiv preprint arXiv:2604.23435*.
- **Khatim, N. A.**, & Arief, M. (2026). Optimized Human-Robot Co-Dispatch Planning for Petro-Site Surveillance under Varying Criticalities. *arXiv preprint arXiv:2602.07924*.
- **Khatim, N. A.**, Irfan, A. A., & Arief, M. M. (2024). Using LLM for Real-Time Transcription and Summarization of Doctor-Patient Interactions into ePuskesmas in Indonesia: A Proof-of-Concept Study. *arXiv preprint arXiv:2409.17054*.
- **Khatim, N. A.**, Irfan, A. A., Hayah, A. M. U., & Arief, M. M. (2024, April). Toward an Integrated Decision-Making Framework for Optimized Stroke Diagnosis with DSA and Treatment under Uncertainty. *In Proceedings of the 2024 6th International Conference on Intelligent Medicine and Image Processing* (pp. 41-49).
- **Khatim, N. A.** (2023). 1D-CNN Implementation for Automatic Epilepsy Detection in EEG Signals based on Interictal Epileptiform Discharge (IED). (*Undergraduate Thesis, Institut Teknologi Sepuluh Nopember*).

HONORS AND AWARDS

1st Place Best Paper Presentation Award

Sep 2024

International Meeting on Medical Network (IMMaN) 2024 Conference

Research Assistance for Research and Innovation Talents Awardee

Jan 2023

National Research and Innovation Agency (BRIN) Indonesia

PRESENTATIONS

Poster Paper Presentation

Jun 2026

The 36th Annual International Council on Systems Engineering (INCOSE) International Symposium, Yokohama, Japan

Oral Paper Presentation

Nov 2025

The INTERSECT Main Conference, Stanford University, USA

Oral Paper Presentation

Sep 2024

The 6th International Meeting on Medical Network (IMMaN) Conference, UIN Jakarta, Indonesia

EXTRACURRICULAR ACTIVITIES

Program Committee

Oct 2025 – Nov 2025

INTERSECT 2025

- Served as an invited judge for the high school research paper presentation track, evaluating submissions and providing structured feedback to student presenters

Student Managerial Trainer

Sep 2021 – Jul 2023

Student Executive Board, Faculty of Intelligent Electrical and Informatics Technology (ELECTICS) ITS

- Delivered and became speaker on self management and event management training sessions

Co-Head of Student Resources Development

Mar 2022 – Jan 2023

Computer-Informatics Engineering Student Association (HMTC) ITS

- Conducted orientation activities, organized meetings, and facilitated regeneration forums within the student association

Co-Head of Organization

Mar 2022 – Jan 2023

Muslim Computer-Informatics Engineering Student Association (KMI) ITS

- Monitored activity progress and scheduled evaluations, ensuring efficient operations

Contest Problem Setter

Apr 2021 – Oct 2021

Schematics National Programming Contest (NPC) ITS

- Created contest problems and managed final round proctoring, ensuring smooth contest operations

Programming Instructor

Sep 2021 – Sep 2021

HMTC Goes To School

- Delivered C/C++ programming language instruction to 25 participants of high school students, designed practical assignments, and facilitated learning

Partnerships and Programs Team

Oct 2021 – Jul 2022

Google Developer Student Club (GDSC) ITS

- Organized club activities and events, collaborating with partners to enhance member engagement

Head of Committee

Oct 2020 – Dec 2020

FesTCival 2020 Informatics ITS

- Led end-to-end planning and execution of the event, coordinating all section teams from pre-event preparation through live operations to post-event wrap-up
- Managed cross-functional coordination across multiple committee divisions to ensure smooth delivery

TECHNICAL SKILLS AND TOOLS

- Programming Languages: C#, C++, Python, Julia, Kotlin, Golang, Typescript
- Web Development: ReactJS, Symfony PHP, .NET Framework, Django REST Framework
- Database Management: PostgreSQL, MySQL
- AI and Machine Learning: PyTorch, TensorFlow, Scikit-learn, POMDPs.jl
- AI-Assisted Development: Claude Code, Codex, Antigravity
- Other: Git, Linux, Docker, Redis, GCP, AWS, HuggingFace